

Assignment – 18:

Problem Statement:

Write a C program that accepts a string from the user and performs the following string operations: i) Calculate length of string ii) String reversal iii) Equality check of two strings iv) Check palindrome v) Check substring.

Code:

```
#include <stdio.h>

#include <string.h>

int main()
{
    char str1[100], str2[100], temp[100];
    printf("Enter a string: ");
    scanf("%s", str1);
    printf("Enter another string for equality check: ");
    scanf("%s", str2);
    printf("Length of string = %lu\n", strlen(str1));
    strcpy(temp, str1);
    int len = strlen(temp);
    for (int i = 0; i < len / 2; i++)
    {
        char ch = temp[i];
        temp[i] = temp[len - i - 1];
        temp[len - i - 1] = ch;
    }
    printf("Reversed string = %s\n", temp);
    if (strcmp(str1, temp) == 0)
    {
        printf("The string is a Palindrome\n");
    }
}
```

```
else
{
    printf("The string is not a Palindrome\n");
}
if (strcmp(str1, str2) == 0)
{
    printf("Strings are equal\n");
}
else
{
    printf("Strings are not equal\n");
}
if (strstr(str1, str2) != NULL)
{
    printf("Substring found in main string\n");
}
else
{
    printf("Substring not found in main string\n");
}
}
```

Input & Output:

```
Enter a string: madam
Enter another string for equality check: mad
Length of string = 5
Reversed string = madam
The string is a Palindrome
Strings are not equal
Substring found in main string
```

```
Enter a string: Hello
Enter another string for equality check: Hell
Length of string = 5
Reversed string = olleH
The string is not a Palindrome
Strings are not equal
Substring found in main string
```

